

**Multiple Choice:**

1. At very high temperatures the extrinsic semiconductors become intrinsic because
 - A. drive in diffusion of dopants and carriers
 - B. band to band transition dominants over impurity ionization
 - C. impurity ionization dominants over band to band transition
 - D. band to band transition is balanced by impurity ionization
2. To forward-bias a diode,
 - A. the anode voltage must be positive with respect to its cathode.
 - B. the anode voltage must be negative with respect to its cathode.
 - C. the cathode voltage must be positive with respect to its anode.
 - D. Either A or B.
3. When a voltage is applied to a semiconductor crystal then the free electrons will flow.
 - A. towards positive terminal
 - B. towards negative terminal
 - C. either towards positive terminal or negative terminal
 - D. towards positive terminal for 1 μ s and towards negative terminal for next 1 μ s
4. A diode is a
 - A. unidirectional device.
 - B. linear device
 - C. nonlinear device
 - D. Both A and C.
5. Avalanche voltage is routinely exceeded when a P-N junction acts as a:
 - A. Current rectifier
 - B. Variable resistor
 - C. Variable capacitor
 - D. Voltage regulator
6. Of the following, which material allows the lowest forward voltage drop in a diode?
 - A. Selenium
 - B. Silicon
 - C. Copper
 - D. Germanium
7. A semiconductor, such as silicon, has an electron valence of
 - A. ± 4 .
 - B. -7 .
 - C. $+1$.
 - D. 0.
8. Assertion (A): The reverse saturation current in a semiconductor diode is 4 nA at 20°C and 32 nA at 50°C. Reason (R): The reverse saturation current in a semiconductor diode doubles for every 10°C rise in temperature.
 - A. Both A and R are true and R is correct explanation of A
 - B. Both A and R are true but R is not a correct explanation of A
 - C. A is true but R is false
 - D. A is false but R is true
9. Determine the diode current at 20 °C for a silicon diode with reverse saturation current of 50 nA and an applied forward bias of 0.6 V.
 - A. 9.17 mA
 - B. 7.19 mA
 - C. 1.79 mA
 - D. 9.7 mA
10. A disadvantage of gallium-arsenide devices is that:
 - A. The charge carriers move fast
 - B. The material does not react to ionizing radiation.
 - C. It is expensive to produce
 - D. It must be used at high frequencies
11. At room temperature the current in an intrinsic semiconductor is due to
 - A. holes
 - B. electrons
 - C. ions
 - D. holes and electrons
12. An *unimportant* factor concerning the frequency at which a P-N junction will work effectively is:
 - A. The type of semiconductor material
 - B. The cross-sectional area of the junction
 - C. The reverse current
 - D. The capacitance with reverse bias
13. Assertion (A): The conductivity of p type semiconductor is higher than that of intrinsic semiconductor. Reason (R): The addition of donor impurity creates additional energy levels below conduction band.
 - A. Both A and R are true and R is correct explanation of A
 - B. Both A and R are true but R is not a correct explanation of A
 - C. A is true but R is false
 - D. A is false but R is true
14. If the reverse bias exceeds the avalanche voltage in a P-N junction:
 - A. The junction will be destroyed
 - B. The junction will insulate; no current will flow
 - C. The junction will conduct current
 - D. The capacitance will become extremely high
15. When a P-N junction does not conduct, it is:
 - A. Reverse biased
 - B. Forward biased
 - C. Biased past the break over voltage
 - D. In a state of avalanche effect
16. When a diode is forward-biased, the anode:
 - A. Is negative relative to the cathode
 - B. Is positive relative to the cathode
 - C. Is at the same voltage as the cathode
 - D. None of the above



17. The sharing of valence electrons in a silicon crystal is called
- A. doping.
 - B. the avalanche effect.
 - C. covalent bonding.
 - D. coupling.
18. The purpose of the I layer in a PIN diode is to:
- A. Minimize the diode capacitance
 - B. Optimize the avalanche voltage
 - C. Reduce the forward break over voltage
 - D. Increase the current through the diode
19. When a silicon diode is conducting at a temperature of 27 °C, reverse saturation current is 35 nA, what is the new reverse current at 50°C?
- A. 175 nA
 - B. 77 nA
 - C. 88 nA
 - D. 23 nA
20. To a first approximation, a forward-biased diode is treated like a (n)
- A. open switch with infinite resistance.
 - B. closed switch with a voltage drop of 0 V.
 - C. closed switch in series with a battery voltage of 0.7 V.
 - D. closed switch in series with a small resistance and a battery.
21. The capacitance of a varactor varies with:
- A. Forward voltage
 - B. Reverse voltage
 - C. Avalanche voltage
 - D. Forward break over voltage
22. A photodiode, when not used as a photovoltaic cell, has:
- A. Reverse bias
 - B. No bias
 - C. Forward bias
 - D. Negative resistance
23. Zener voltage is also known as:
- A. Forward break over voltage
 - B. Peak forward voltage
 - C. Avalanche voltage
 - D. Reverse bias
24. An n-type semiconductor is a semiconductor that has been doped with
- A. trivalent impurity atoms.
 - B. impurity atoms whose electron valence is + 4.
 - C. pentavalent impurity atoms.
 - D. None of the above.
25. The purpose of doping is to:
- A. Make the charge carriers move faster
 - B. Cause holes to flow
 - C. Give a semiconductor material certain properties
 - D. Protect devices from damage in case of transients
26. Which of the following *does not* result from adding an acceptor impurity?
- A. The material becomes P type
 - B. Current flows mainly in the form of holes
 - C. Most of the carriers have positive electric charge
 - D. The substance has an electron surplus
27. A pure semiconductor is often referred to as a (n)
- A. extrinsic semiconductor.
 - B. intrinsic semiconductor.
 - C. doped semiconductor.
 - D. None of the above.
28. If ac is applied to a diode, and the peak ac voltage never exceeds the avalanche voltage, then the output is:
- A. AC with half the frequency of the input
 - B. AC with the same frequency as the input
 - C. AC with twice the frequency of the input
 - D. None of the above
29. When a P-N junction is reverse-biased, the capacitance depends on all of the following *except*:
- A. The frequency
 - B. The width of the depletion region.
 - C. The cross-sectional area of the junction
 - D. The type of semiconductor material
30. If an electron has a charge of -1 unit, a hole has:
- A. A charge of -1 unit
 - B. No charge
 - C. A charge of +1 unit
 - D. A charge that depends on the semiconductor type
31. Holes flow the opposite way from electrons because:
- A. Charge carriers flow continuously
 - B. Charge carriers are passed from atom to atom
 - C. They have the same polarity
 - D. Holes flow in the same direction as electrons
32. In an n-type semiconductor, the minority current carriers are
- A. free electrons.
 - B. protons.
 - C. valence electrons.
 - D. holes.
33. When used as a voltage regulator, a zener diode is normally
- A. forward-biased.
 - B. reverse-biased.
 - C. not biased.
 - D. None of the above.
34. The most commonly used semiconductor material is
- A. silicon
 - B. germanium
 - C. mixture of silicon and germanium
 - D. none of the above



35. A crystal set:

- A. Can be used to transmit radio signals
- B. Requires a battery with long life
- C. Requires no battery
- D. Is useful for rectifying 60-Hz ac

36. The forward break over voltage of a silicon diode is:

- A. About 0.3 V
- B. About 0.6 V
- C. About 1.0 V
- D. Dependent on the method of manufacture

37. Which of the following is *not* an advantage of semiconductor devices over vacuum tubes?

- A. Smaller size.
- B. Lower working voltage.
- C. Lighter weight.
- D. Ability to withstand high voltages.

38. A neutral atom with an atomic number of three has how many electrons?

- A. 1
- B. 3
- C. none
- D. depends on the type of atom

39. Which of these diode types might be found in the oscillator circuit of a microwave radio transmitter?

- A. A rectifier diode
- B. A cat whisker
- C. An IMPATT diode
- D. None of the above

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- C. pentavalent impurity atoms.
- D. None of the above.

41. A diode detector:

- A. Is used in power supplies

B. Is employed in some radio receivers

- C. Is used commonly in high-power radio transmitters
- D. Changes dc into ac

42. Selenium works especially well in:

- A. Photocells
- B. High-frequency detectors
- C. Radio-frequency power amplifiers
- D. Voltage regulators

43. A semiconductor material is made into N type by:

- A. Adding an acceptor impurity
- B. Adding a donor impurity
- C. Injecting electrons
- D. Taking electrons away

44. The charge on a single electron is

- A. $6.25 \times 10^{-18} \text{ C}$
- B. $1.6 \times 10^{-19} \text{ C}$
- C. $1.6 \times 10^{-21} \text{ C}$
- D. $3.14 \times 10^{-6} \text{ C}$

45. Holes flow from:

- A. Minus to plus
- B. Plus to minus
- C. P-type to N-type material
- D. N-type to P-type material

46. A diode audio limiter circuit:

- A. Is useful for voltage regulation
- B. Always uses Zener diodes
- C. Rectifies the audio to reduce distortion
- D. Can cause objectionable signal distortion

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- A. free electrons.
- B. valence electrons.
- C. protons.
- D. holes.

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49. Determine the temperature coefficient of a 5 V zener diode (rated 25°C value) if the nominal voltage drops to 4.8 V at a temperature of 100°C.

- A. 0.048 %/°C
- B. 0.053 %/°C
- C. 0.064 %/°C
- D. 0.07 %/°C

50. The term “semiconductor” arises from:

- A. Resistor-like properties of metal oxides.
- B. Variable conductive properties of some materials.
- C. The fact that there’s nothing better to call silicon.
- D. Insulating properties of silicon and GaAs.

END